ECE 310 Fall 2026
Lecture 10: Transfer Functions and LTI System Response: Part 2

Corey Snyder

University of Illinois Urbana-Champaign

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Learning Objectives

- Work with transfer functions given as improper rational expressions.
- Use z-transform properties and system algebra to compute the response and transfer function of larger LTI systems composed of multiple LTI systems.



Improper Rational Expressions

Thus far, we have only considered transfer functions described by proper rational expressions:

$$H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}, \; M \leq N.$$

If $M > N$, we will have an *improper rational expression*. In this case, we will have two options to obtain the inverse z-transform.



Inverse z-transform with Improper Rational Expressions

First, let's see when improper rational expressions may appear. Consider the following causal LCCDE:

$$y[n] = 2y[n-1] + 3y[n-2] + x[n] - 3x[n-1] + x[n-2] + 4x[n-3].$$

To begin, we will determine the transfer function $H(z)$ for this system.



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$$Y(z) = 2Y(z)z^{-1} + 3Y(z)z^{-2} + X(z) - 3X(z)z^{-1} + X(z)z^{-2} + 4X(z)z^{-3}$$



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$$Y(z)(1 - 2z^{-1} - 3z^{-2}) = X(z)(1 - 3z^{-1} + z^{-2} + 4z^{-3})$$

$$\frac{Y(z)}{X(z)} = H(z) = \frac{1 - 3z^{-1} + z^{-2} + 4z^{-3}}{1 - 2z^{-1} - 3z^{-2}}$$

$M = 4$ inputs

$N = 2$ feedback terms

Approach #1: Time-shifting and Linearity

What if we let the numerator be 1?

$$H(z) = \frac{1 - 3z^{-1} + z^{-2} + 4z^{-3}}{1 - 2z^{-1} - 3z^{-2}} = (1 - 3z^{-1} + z^{-2} + 4z^{-3}) \underbrace{\frac{1}{1 - 2z^{-1} - 3z^{-2}}}_{G(z)}$$

$G(z) = \frac{1}{1-2z^{-1}-3z^{-2}} = \frac{1}{(1-3z^{-1})(1+z^{-1})}$

$H(z) = G(z) - 3z^{-1}G(z) + z^{-2}G(z) + 4z^{-3}G(z)$

$\downarrow \mathcal{Z}^{-1}$

$h[n] = g[n] - 3g[n-1] + g[n-2] + 4g[n-3]$

$G(z) = \frac{A_1}{1-3z^{-1}} + \frac{A_2}{1+z^{-1}}$

$\downarrow$ PFD

$G(z) = \frac{\frac{3}{4}}{1-3z^{-1}} + \frac{\frac{1}{4}}{1+z^{-1}}$

$\mathcal{Z}^{-1} \rightarrow g[n] = \frac{3}{4}(3)^n u[n] + \frac{1}{4}(-1)^n u[n]$

$a^n u[n] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}}$, $|z| > |a|$

Approach #2: Polynomial Long-division

If $H(z)$ is an *improper rational expression*, the transfer function may be written as

$$H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \sum_{k=0}^{M-N-1} C_k z^{-k} + \frac{\sum_{k=0}^{N-1} d_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}}$$

where $M > N$.

To find the inverse z-transform, we can follow a two-step process:

1. Determine the C_k coefficients by performing polynomial long division between the original numerator $B(z)$ and denominator $A(z)$, i.e. $B(z)/A(z)$.
2. Perform partial fraction decomposition on the remainder of polynomial long division to determine the remaining A_k coefficients.

Altogether, this procedure will give us

$$H(z) = \sum_{k=0}^{M-N-1} C_k z^{-k} + \sum_{k=1}^{N} \frac{A_k}{1 - p_k z^{-1}}.$$



Inverse z-transform with Improper Rational Expressions

$$y[n] = 2y[n-1] + 3y[n-2] + x[n] - 3x[n-1] + x[n-2] + 4x[n-3]$$

$$H(z) = \frac{1 - 3z^{-1} + z^{-2} + 4z^{-3}}{1 - 2z^{-1} - 3z^{-2}} = \sum_{k=0}^{M-N-1} C_k z^{-k} + \sum_{k=1}^{N} \frac{A_k}{1 - p_k z^{-1}}$$

First, we will determine the C_k coefficients by performing polynomial long division.

$$\begin{array}{r} \frac{5}{9} - \frac{4}{3}z^{-1} \\ 1 - 2z^{-1} - 3z^{-2} \overline{)\ 1 - 3z^{-1} + z^{-2} + 4z^{-3}} \\ -\ \ -\frac{4}{3}z^{-1} + \frac{8}{3}z^{-2} + 4z^{-3} \\ \hline 1 - \frac{5}{3}z^{-1} - \frac{5}{3}z^{-2} \\ -\frac{5}{9} - \frac{10}{9}z^{-1} - \frac{5}{3}z^{-2} \\ \hline \frac{4}{9} - \frac{5}{9}z^{-1} \end{array}$$

$M = 4$ inputs
$N = 2$ feedback terms

$$H(z) = \frac{5}{9} - \frac{4}{3}z^{-1} + \frac{\frac{4}{9} - \frac{5}{9}z^{-1}}{1 - 2z^{-1} - 3z^{-2}}$$

C_0 C_1

Inverse z-transform with Improper Rational Expressions

$$y[n] = 2y[n-1] + 3y[n-2] + x[n] - 3x[n-1] + x[n-2] + 4x[n-3]$$

$$H(z) = \frac{5}{9} - \frac{4}{3}z^{-1} + \frac{\frac{4}{9} - \frac{5}{9}z^{-1}}{1 - 2z^{-1} - 3z^{-2}} = \frac{5}{9} - \frac{4}{3}z^{-1} + \frac{A_1}{1 - 3z^{-1}} + \frac{A_2}{1 + z^{-1}}.$$

Finally, we can find A_1 and A_2 by partial fraction decomposition and determine $h[n]$.

① $\frac{\frac{4}{9} - \frac{5}{9}z^{-1}}{1 - 2z^{-1} - 3z^{-2}} = \frac{\frac{4}{9} - \frac{5}{9}z^{-1}}{(1 - 3z^{-1})(1 + z^{-1})} = \frac{A_1}{1 - 3z^{-1}} + \frac{A_2}{1 + z^{-1}}$ PFD $\rightarrow = \frac{\frac{7}{36}}{1 - 3z^{-1}} + \frac{\frac{1}{4}}{1 + z^{-1}}$

$h[n] = \frac{5}{9}\delta[n] - \frac{4}{3}\delta[n-1] + \frac{7}{36}(3)^n u[n] + \frac{1}{4}(-1)^n u[n]$

② $\frac{4}{9} - \frac{5}{9}z^{-1} = A_1(1 + z^{-1}) + A_2(1 - 3z^{-1})$

③ A_1: $z = 3$

$\frac{4}{9} - \frac{5}{9}(3)^{-1} = A_1(1 + (3)^{-1}) \rightarrow A_1 = \frac{7}{36}$

A_2: $z = -1$

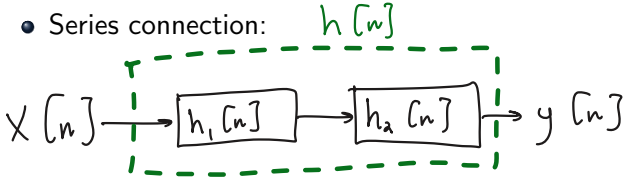
$\frac{4}{9} - \frac{5}{9}(-1)^{-1} = A_2(1 - 3(-1)^{-1}) \rightarrow A_2 = \frac{1}{4}$

Equivalent to Approach #1!

System Algebra

System algebra concerns the connection of multiple LTI systems to form a larger LTI system. Each connection of two systems will be in either series of parallel:

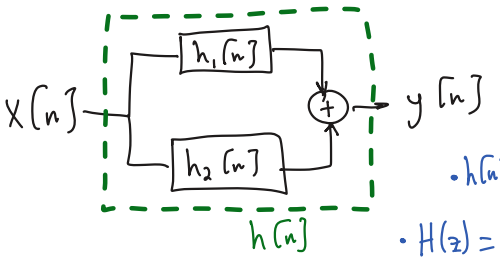
- Series connection:



$$y[n] = x[n] * h[n]$$
$$= x[n] * (h_1[n] * h_2[n])$$

- $h[n] = h_1[n] * h_2[n]$
- $\updownarrow \mathcal{Z}$
- $H(z) = H_1(z) H_2(z)$

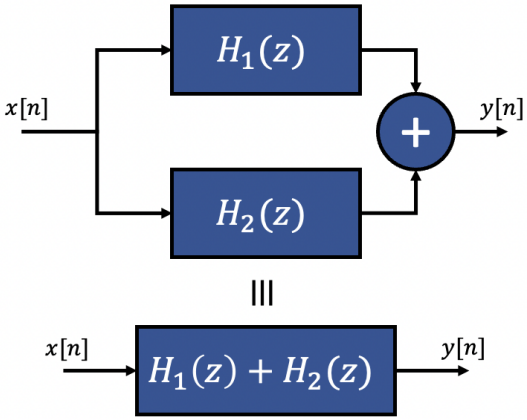
- Parallel connection:

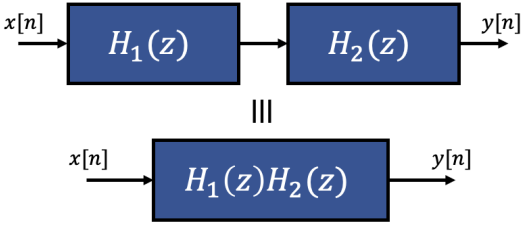


- $h[n] = h_1[n] + h_2[n]$
- $\updownarrow \mathcal{Z}$
- $H(z) = H_1(z) + H_2(z)$

$$y[n] = x[n] * h[n]$$
$$= x[n] * h_1[n] + x[n] * h_2[n]$$
$$= x[n] * (h_1[n] + h_2[n])$$

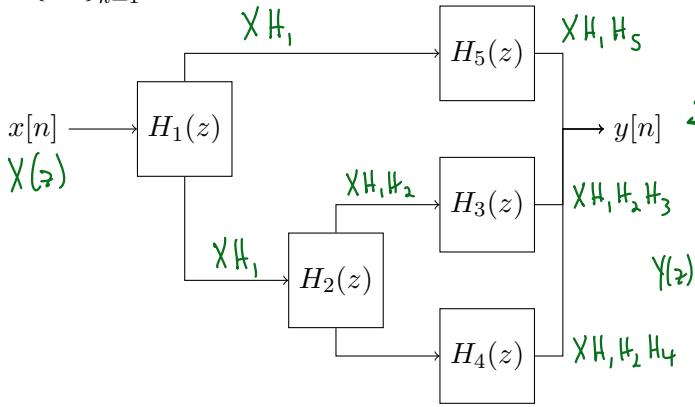
System Algebra





System Algebra Example

Determine the frequency response $H(z)$ of the below overall system in terms of each of $\{H_k\}_{k=1}^{5}$.



$\xrightarrow{\mathcal{Z}}$ $Y(z) = X(z)\big(H_1(z)H_5(z) + H_1(z)H_2(z)H_3(z) + H_1(z)H_2(z)H_4(z)\big)$

$Y(z) = X(z)\,H_1(z)\big(H_5(z) + H_2(z)\big(H_3(z)+H_4(z)\big)\big)$

$\dfrac{Y(z)}{X(z)} = H(z)$

$H(z) = H_1(z)\big(H_5(z) + H_2(z)\big(H_3(z)+H_4(z)\big)\big)$